

```
# 1. Take 5 inputs from the user
user_list = []
print("Enter 5 values:")
for i in range(5):
    val = input(f"Value {i + 1}: ")
    user_list.append(val)

# 2. Display all elements
print("\nElements in the list:")
for item in user_list:
    print(item, end=" ")
print()

# 3. Calculate and display hash code
# Note: Lists are mutable, so we convert to a tuple to obtain the hash
code
list_hash = hash(tuple(user_list))
print(f"Hash code of the list: {list_hash}")
```

```
... Enter 5 values:
Value 1: 55
Value 2: 554
Value 3: 5656
Value 4: 654
Value 5: 55

Elements in the list:
55 554 5656 654 55
Hash code of the list: -2140959073547684158
```

```
import hashlib

class Block:
    def __init__(self, data, previous_hash="0"):
        self.data = data
        self.previousHash = previous_hash
        self.hash = self.calculateHash()

    def calculateHash(self):
        # Concatenate block contents to form the input string
        block_content = f"{self.data}{self.previousHash}"
        return hashlib.sha256(block_content.encode('utf-8')).hexdigest()
```

```
# Creating 3 blocks and linking them
# Block 1 (Genesis Block)
block1 = Block(data="Genesis Block", previous_hash="0")

# Block 2 (Linked to Block 1)
block2 = Block(data="Transaction: Alice -> Bob 10 BTC",
previous_hash=block1.hash)

# Block 3 (Linked to Block 2)
block3 = Block(data="Transaction: Bob -> Charlie 5 BTC",
previous_hash=block2.hash)

# Storing in a list to display relationships
blockchain = [block1, block2, block3]

for index, block in enumerate(blockchain, start=1):
    print(f"--- Block {index} ---")
    print(f"Block Data:      {block.data}")
    print(f"Previous Hash:   {block.previousHash}")
    print(f"Current Hash:     {block.hash}")
    print(f"Hash Length:      {len(block.hash)} characters\n")
```

```
... --- Block 1 ---
Block Data:      Genesis Block
Previous Hash:   0
Current Hash:    8500b59bb5271135cd9bcbf0afd693028d76df3b9c7da58d412b13fc8a8f9394
Hash Length:    64 characters

--- Block 2 ---
Block Data:      Transaction: Alice -> Bob 10 BTC
Previous Hash:   8500b59bb5271135cd9bcbf0afd693028d76df3b9c7da58d412b13fc8a8f9394
Current Hash:    6b3c37bfe1alc96bae092af9ea71f3892706fc19739f14a79a8bc5875287199e
Hash Length:    64 characters

--- Block 3 ---
Block Data:      Transaction: Bob -> Charlie 5 BTC
Previous Hash:   6b3c37bfe1alc96bae092af9ea71f3892706fc19739f14a79a8bc5875287199e
Current Hash:    7eb7ec1f158fd2bddfe13d34fb78d7d9e011355fe10454bc5c20097d7b383359
Hash Length:    64 characters
```